



Entanglement is an extremely strong correlation that exists between quantum particles—so strong that two or more quantum particles can be inextricably linked in perfect unison, even if separated by great distances. The particles remain perfectly correlated even if separated by great distances. These are so intrinsically connected that these can be said to dance in instantaneous, perfect unison, even when placed at opposite ends of the universe.

Such quantum effects are extremely useful to the future of computing and communications technology. Thanks to superposition and entanglement, a quantum computer can process a vast number of calculations simultaneously. Where a classical computer works with 0s and 1s, a quantum computer will have the advantage of using 1s, 0s and superpositions of 1s and 0s.

Current status

Qubits could be made of photons, atoms, electrons, molecules or perhaps something else. But these are notoriously tricky to manipulate, since any disturbance causes these to fall out of their quantum state (or decohere).

Decoherence is the Achilles heel of quantum computing, but it is not insurmountable. The field of quantum error correction examines how to stave off decoherence and combat other errors. While quantum computers have been theoretically demonstrated to have incredible potential, and scientists are working around the world to realise that potential, there is much work to be done before these hit the market.

There are quantum computers already, but not of sufficient power to replace classical computers. While practical quantum technologies are already emerging—including highly-effective sensors, actuators and other devices—a true quantum computer that outperforms a classical computer is still years away.

Theorists are continually figuring

out better ways to overcome decoherence, while experimentalists are gaining more and more control over the quantum world through various technologies and instruments. Pioneering work being done today is paving the way for the upcoming quantum era.

Potential applications

Quantum computers will be able to efficiently simulate quantum systems. This will allow to study, in remarkable detail, interactions between atoms and molecules. This could help design new drugs and materials, such as superconductors that work at room temperature.

Another of the many benefits of quantum computers over classical ones is searching through a space of potential solutions for the best solution. Researchers are constantly working on new quantum algorithms and applications. But the true potential of quantum computers likely has not even been imagined yet.

Future uses of quantum computers are bound only by imagination. Quantum technologies offer ultra-secure communications, sensors of unprecedented precision and computers that are exponentially more powerful than any supercomputer for a given task. These technologies are destined to fundamentally change our lives, and the first commercially-available quantum devices are only now beginning to emerge.

Quantum computing has the capability to unlock answers to some of humanity's most pressing questions that are presently unsolvable with current computing technologies. It is expected that in less than ten years, quantum computers will begin to outperform everyday computers, leading to breakthroughs in artificial intelligence, discovery of new pharmaceuticals and beyond.

The very fast computing power of quantum computers has the potential to disrupt traditional businesses and challenge cyber security.

Businesses need to be ready for a quantum future because it is coming. The technology could herald radical changes for the following areas, to name a few:

- Safer airplanes. Lockheed Martin plans to use its D-Wave to test jet software that is currently too complex for classical computers.
- Discover distant planets. Quantum computers will be able to analyse the vast amount of data collected by telescopes and seek out Earth-like planets.
- Win elections. Campaigners will comb through reams of marketing information to best exploit individual voter preferences.
- Boost GDP. Hyper-personalised advertising, based on quantum computation, will stimulate consumer spending.
- Detect cancer earlier. Computational models will help determine how diseases develop.
- Help automobiles drive themselves. Google is already using a quantum computer to design software that can distinguish cars from landmarks.
- Reduce weather-related deaths. Precision forecasting will give people more time to take cover.
- Cut back on travel time. Sophisticated analysis of traffic patterns in the air and on the ground will forestall bottlenecks and snarls.
- Develop more effective drugs. By mapping amino acids, for example, or analysing DNA-sequencing data, doctors would be able to discover and design superior drug based treatments.

Conclusion

Developed countries are making huge investments for the development of quantum technologies in order to become the epicentres of this technology revolution in the near future. However, quantum computing might struggle to impact everyday life as it may be suppressed by those opposed to the changes it might bring. **EFY**